SPEC OPS POWER

How Elite Special Forces Commandos and Intelligence Operatives Stay Powered-Up On Top Secret Operations

By ***New York Times*** bestselling author
Jason Hanson With **Steven Harris**

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Published by Spy Briefing, LLC

124 S. Main Street #4090
Cedar City, Utah 84720

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Introduction

Welcome to *Spec Ops Power*. This will become your Bible in how to have a simple system to power your critical devices in a blackout scenario.

There is a tremendous amount of material covered here, but you will soon see the simplicity in this system. The fact is, our elite warriors need simplicity when they find themselves in the rougher areas of the world. This battery system does the job for them and it will do the job for you too.

I highly encourage you to not only read this book but, of course, to go out and quickly build your own battery power system so that you and your family are prepared when the lights go out in your own home.

Now, let's get down to business and get you better prepared for the days ahead. Please note, we go very in-depth and you can skip around from section to section if you want to build your system quickly. Let's get started…

The picture below shows a thorough battery power system that's laid out on a kitchen table.

Don't worry, I am going to break it down and show you what's going on and how easy it is to do this.

As you can see, I am powering an iPhone, an iPad, an Android, an android Pad, a radio, a scanner, an AA charger and various other things all at the same time.

When the power fails, all you have to do is connect up your inverter, your lights, your radio, your AA chargers and away you go.

But, there are a lot of little details that go into making a battery power system. So, before I start, we are going to help you select which battery is right for you.

For the purpose of this book, we are going to exclusively deal with lead acid batteries. There is just no competition to these in capacity, price and ease of use. Sorry, no nickel iron, no nickel metal hydride and no lithium. They all pale in comparison to the availability, the price, the amount of energy, the chargers and the tools you can get to work with this very proven technology.

The scientific area of lead acid batteries is complicated, that's why I made this book for you. Just the subject of one type of lead acid battery can fill several large volumes of books. I have most of these books in my personal library, so what I am going to do is teach you what generally applies to you.

I'm going to teach you the rules of thumb. Information that you can remember and use and make very smart decisions with.

SAFETY

Let’s begin with some safety concerns regarding lead acid batteries. They contain sulfuric acid, it will blind you, it will eat through your clothes, it'll burn your skin, it will kill you if you drink it.

We are going to talk about acid safety procedures, wearing gloves and eyeglasses and when you need to do this. We're also going to talk about acid safe batteries that have no way of ever leaking any sulfuric acid.

When we talk about testing the acid density to determine the level charge in a battery, we'll talk more about safety glasses and gloves.

If you short a cable or a wrench or anything else conductive crosses the battery terminals, it'll spark like crazy and it will probably glow red from too much current flowing through it and it will cause a fire, so do not short out the batteries.

We will also address the very, very overblown issue of hydrogen generation. Of hydrogen being made while the battery is being charged. However, first, we need to speak the same language back and forth. I want to teach you how to speak battery.

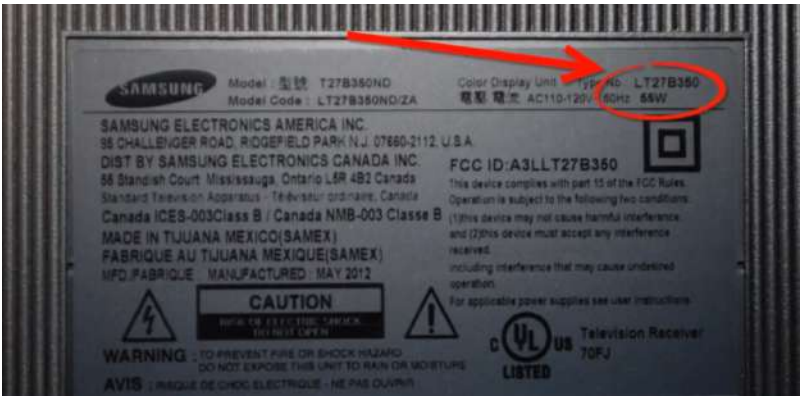
TERMINOLOGY

Lead acid batteries are electrochemical storage devices for electricity. Basically, we have volts, amps, ampere hours (Ah) which we'll talk about in detail, watts and we'll mention those, watt hours (Wh), ohms, and even more nomenclature.

I'm not going to make you learn all of these. I am just going to talk to you about watts and ampere hours and a little bit about volts and amps. Now, I can't tell you what is the best battery for you, you have to make that determination. But, I can help you make that decision and I guarantee you I will, it's pretty easy and straightforward.

If you look at the back of most items that run on 120 volts AC, which is what comes out of your wall in your house, it will tell you how much power it uses in the unit watts.

For example: I have a 28-inch Samsung TV monitor and on the back of this, the label says it uses 55 watts, that's it. Now, the TV uses 55 watts, what are some wattages of stuff around your house?



- Light bulbs range from 2 watts to 15 watts.
- Compact fluorescent light bulbs, still one of my favorites, can be 5, 7, 13 to 23 watts.
- Incandescent light bulbs, the old-style, get very hot, 25, 40, 60, 75, 100, 200 and 300 watts were the common sizes.
- An iPhone while charging uses 5 watts.

- An iPad uses 10 watts.
- My 28-inch LCD TV was 55, as I said.
- A new Energy Star rated refrigerator will draw only between 120 and 250 watts when it's running.

Now, a coffee maker, the thing that a lot of people want to plug in first thing in the morning, is going to be 1100 to 1200 watts. Your hairdryer is 1500 watts. Your microwave is 1500 watts.

So, how long can we power these off of a battery system? To know that number, we need to know the voltage. 12 volts comes from the battery power system and goes to the back of the inverter. We will always be using 12 as our number of volts.

Okay, please bear with me, here is a little math that I'll try to make straightforward and easy, okay? Let's say you were using six 4-watt LED light bulbs to light up your house from a battery.

So, you got 6 bulbs that draw 4 watts each, 6 x 4 equals 24. You take 24 watts and you divide it by 12 volts. We're going to use 12 volts throughout the entire book because our battery is 12 volts. You divide it by 12 volts, you get 2 amps. Watts divided by volts is amps.

Lets say you are using 6 four watt LED light Bulbs to light up your house from a battery.

Watts 24 watts
------- = amps ---------- = 2 amps
volts 12 volts

So if you leave the lights on for 1 hour you will draw 2 amps for an hour. This is 2 ampere hours.

Ampere Hours are abrevieted Ah

If you left 2 amps on for 10 hours, that would be 20 ampere hours. 2 amps x 10 hours = 20 Ah

So, if you leave the lights on for 1 hour, you will draw 2 amps for an hour, that is 2 ampere hours. Ampere hours are abbreviated with Ah. You'll see that through the rest of the book.

If you left 2 amps on for 10 hours, that would be 20 Ah, 2 amps times 10 hours equals 20 Ah. 10 amperes for 400 hours would be 100 Ah. 20 amperes for 10 hours will be 200 Ah. You just multiply the two together and you get the number of ampere hours.

How much energy or power does a battery have? How long will it last? This is the

biggest question that I get. How long will it last?

The batteries are rated in ampere hours (Ah), the largest typical marine battery from Sam's or Walmart is rated around 100 Ah. This would be called a group 31 battery. So, if you're drawing 2 amps for 1 hour, that's a 2 Ah draw on the battery per hour.

100 Ah / 2 Ah = 50 hours of operation. (roughly)

Those 6 - 4 Watt LED Bulbs would stay on for about 50 hours. That is a good 6+ Days with 8 Hours a day of use.

I'll give you another example. A 1200-watt coffee maker divided by 12 volts from the battery equals 100 amps per hour. A 100 Ah battery divided by 100 amps per hour equals one hour of operation.

But, in reality, it probably only lasts 30 minutes because when you pull too many amps too fast off a battery, you have chemical inefficiencies and you don't get all of the power out.

1200 watt coffee maker
---------- = 100 amps per hour
12 volt battery

100 Ah battery / 100 amps per hour = 1 hour of operation

Full battery life happens when you draw small amounts of current from a battery, like around 1/20 of its ability. So, a 100 Ah having 5 amps pulled from it will last for the full Ah.

Plainly, 100 Ah divided by 5 ampere per hour equals 20 hours, straightforward math. 5 amps times 12 volts equals 60 watts. That's a lot of 2 or 4 watt LED bulbs. That's between 30 and 15 light bulbs.

100 Ah
---------- = 20 Hours
5 amps / hour

5 amps x 12 volts = 60 watts

Quite a few phones can be charged at 5 watts a piece too. At the much higher current, like 20, 50 or 100 amps, you get into the chemical and inefficiencies of the battery.

For instance, if you pour 5 gallons of water out of a 5-gallon bucket in 1 second, you get 5 gallons. But, if you try to pull 100 amps out of a 100 Ah battery, you'll get like 100 amps for 30 minutes rather than 60. It's like pouring the five gallons of water into the other bucket so quickly, that some of it is going to splash out on you.

When you are on battery power, as I say, think like a submarine below the surface. Run silent, run deep and run on low power loads.

Now, let's put this battery into a little better perspective for you.

If you have a 12 volt 100Ah Battery that is 12 x 100ah = 1200 watt hours.

A small $160 2 cycle 1000 watt generator will put out 500 watts of electricity for 8.5 hours on 1.2 gallons of fuel.

500 watts x 8.5 hours = 4250 watt hours on 1.2 gallons of gasoline.

Your battery has 1200-watt hours in it, so 1200 Watts divided by 4250 watts equals 0.28, 0.28 times 1.2 gallons of gasoline, that the generator used, equals 0.34 gallons of gasoline or about 43 ounces, just more than a quart.

$$\frac{1200 \text{ watts}}{4250 \text{ watts}} = 0.28$$

0.28 x 1.2 gallons gasoline = 0.34 gallons of gasoline

A single large marine battery is around 100 Ah, but most of them, the group 29s, are around 75 Ah. So, imagine for each battery you have that you have about the same power as if you had one quart of gasoline for a generator and then you captured all that electricity coming off the generator while burning that one quart.

So, why do we want to use a battery? Because if you try to run 12 watts of lights off a 1000-watt generator, it won't just use that amount of gasoline to make 12 Watts.

It'll be using about the same amount of gasoline it would to run at one quarter of its output, so around 250 Watts.

It's a generator, it has to maintain a fuel to air ratio, it has a minimum amount of fuel that it needs to keep itself spinning. If you plug in a 12-watt load, it's still going to burn 250 Watts worth of gasoline.

Now, for a battery, if you want 12 watts or 4 watts or 2 watts, it will output just that amount of power that you need. A 2-watt LED nightlight on a battery would last you 25 days if you have it running 24 hours a day.

A 2 watt LED night light on a battery would last
2 watts / 12 volts = 0.166 amps

100 Ah
--------- = 600 hours / 24 hours / day = 25 Days.
0.166 amps

Okay, enough math. Just remember, your big marine battery is equal to about one quart of gasoline in a generator.

Keep that in mind when you go and plug in your coffeemaker or your refrigerator or your freezer, those higher power loads just are not going to last very long on a battery.

Even though you don't want to run your 1200-watt coffee maker off the lead acid battery, you could run your 600+ watt sump pump off of it.

True, it might be drawing 600 watts of power or more, but a sump pump is usually only doing its job from 30 seconds to a minute to 2 minutes and that sure beats hauling water up the steps in a bucket and pouring it outside.

You might only have 1 to 2 hours of sump pumping ability, but you break that down into 1-minute pumping sessions and that's 60 to 120 pumps. If it only fills up once every 15 minutes, then that's 15 to 30 hours of sump pump back up off of one marine battery. If your sump pump is a big concern, you might consider having a larger battery system.

There is one more way of looking at this, kind of a better way, in terms of your iPhone battery. An iPhone 5 has a battery rated at 5.45-watt hours. As I've previously shown, your 100 Ah lead acid battery is about 1,200 watts, so 1,200 divided by 5.45 equals 220. This means your 100 Ah lead acid marine battery could charge your iPhone about 220 times.

$$\frac{1200}{5.45} = 220$$

x 220

So, comparing 100 Ah lead acid battery to gasoline in a generator, it does not look so big, does it? But, comparing it to an iPhone, it looks huge!

BATTERY TYPES

There are 3 main types of lead acid batteries.

One: There is what we call a flooded battery, this is a regular battery like you're used to seeing all the time and one that's in your car.

Two: We have a sealed lead-acid battery, an SLA battery also called a Gel Cell.

Three: An absorbed glass mat battery or an AGM battery. These are the most lead safe batteries you are going to find.

Here's some photo examples of flooded lead-acid batteries. A marine battery, golf cart batteries and big Trojan golf cart batteries.

Also, the starting battery in your car is a flooded lead-acid battery. Flooded lead-acid batteries are the most affordable and are ones that contain a liquid sulfuric acid.

You can open the top of the battery and you can see the battery acid inside of it.

Examples of these would be: Car starting batteries, regular marine batteries and golf cart batteries. All of which can range anywhere from 30 Ah to over 1500 Ah each.

Normally, they come in 12 volts. Although, golf cart batteries come in both 6 volts and 12 volts.

These batteries have to remain standing up, they can bounce around all day in a golf cart, but they do not like to operate on their side or upside down

Also, if you “boil off” part of the electrolyte, the sulfuric acid in water, you can add more distilled water to the battery to bring it back up to its level.

Water must be distilled, like the jugs of distilled water from the grocery store and they work just fine.

A typical car battery in a vehicle can last up to 5 years depending upon the climate that it is in. A typical marine battery stored and used at around 77 degrees Fahrenheit and kept charged when not in use, can last about 5 years.

In hotter climates such as deserts of Arizona, this life can be half as much. Larger lead acid battery such as golf cart batteries, if kept charged when not in use and not abused with over discharge, can last 10 years or more.

The flooded lead acid battery will be rated in either Ampere Hours (Ah) or Reserve Capacity (RC) in minutes. I'll tell you how to convert these numbers in a little bit. Unless you have children and you're concerned with them getting their fingers into the battery acid, you are probably going to be using a flooded lead acid battery.

Our second type of lead acid battery is a sealed lead acid battery, an SLA battery, also called a Gel Cell. Rather than submerging the plates in a liquid, the electrolyte is impregnated into a moistened separator.

These batteries, for example, are used in deer feeders, emergency lights, such as exit signs, they're used in your child's electric vehicles, the pink Barbie Jeep your daughter drives around, electric bikes and razor scooters, to name a few.

There is no danger of spilling acid with an SLA battery or a Gel Cell. These batteries

are sealed and will only vent if they are charged too quickly and get too hot. They are charged at a slower rate to prevent overheating. Gel cells can operate in any position, right side, upside down, on their side, it just doesn't matter.

SLA or gel cells are significantly more expensive than flooded lead acid batteries and usually have a shorter life because their electrolyte cannot be replaced or topped off with distilled water.

These batteries might be one year old by the time they make it here from China, get distributed in the USA and finally make it to a store shelf. Then, if not used properly and kept charged when in storage, they can have a life of only another 1 or 2 years.

If used properly, they can have a life of up to about 5 years. If you were taking a battery into the field or the woods, you'd be taking an SLA or a gel cell battery with you because they conveniently come in sizes such as 6 Ah, 12 Ah and they're like sealed plastic bricks.

SLA batteries can run 2 to 3 times the cost of a flooded lead acid battery. These

batteries are generally rated in Ah and they're clearly marked on the battery as such.

The third type of lead acid batteries are Absorbed Glass Mat, also better known as AGM batteries. The AGM technology was developed in 1985 for military aircraft to reduce weight, increase power handling and improve reliability.

The acid is absorbed in a very fine fiberglass mat, making the battery spill-proof. This enables shipment without hazardous material restrictions. The plates can be made flat to resemble a standard flooded lead acid battery pack in a rectangular case or they can be wound in a cylindrical cell. Optima brand are the ones who are famous for this.

These are the Cadillacs of lead acid batteries. They are sealed and do not leak gas because they recombine the hydrogen and oxygen from charging back in the water, you'll never see it.

These are the only lead acid batteries that are so safe, you can fly with them on a jet legally. If you want to carry a 50-pound lead acid battery onto a plane, this is the one you would do it with.

Like the SLA Gel Cells, these are 100% sealed and your kids or grandkids won't get hurt by any sulfuric acid. Leaving a lead acid battery partially charged or fully uncharged hurts the battery through the formation of sulphate crystals, also called Sulfation.

Whereas, a flooded lead acid battery and the SLA batteries really want to be kept at 100% charge all the time, AGM batteries can go for a longer time before needing to be topped off at 100%.

Like flooded lead acid batteries, AGM come in sizes from about 30 Ah to 105 Ah or larger. They are basically the same size as a marine battery, but exceptions can be found to this.

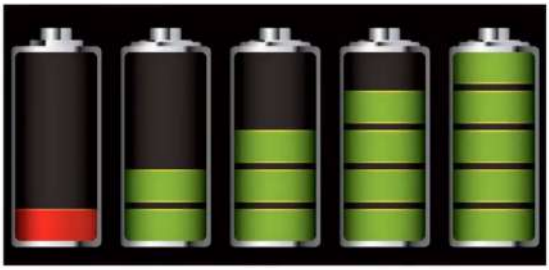
One of the best-known makers of spiral wound AGM lead acid batteries is Optima. AGM batteries will easily sell at double the price of a flooded lead acid battery, like a marine battery.

Double the price is still quite affordable. However, where a marine battery might have a decent life of 5 years, the AGM battery can have an extended life of up to nearly 10 years, if properly maintained and not abused through too heavy of a discharge.

These batteries will be rated in either ampere hours (Ah) or reserve capacity (RC). AGM batteries will be the highest cost, longest life, most dependable and safest battery you will be able to purchase.

KILLING A BATTERY

How to kill a battery? What is depth of discharge? This is something very important that you must know as your battery will not live forever. The use of the battery damages it and eventually it will not work. However, this is a relatively slow process.



If you have a 100% full battery that has been on a computer-controlled 3 stage charger and has been maintained, it's full. Let's say, it's a 100 Ah battery. Take 30 Ah or 30% out of the battery and then recharging it would be a cycle. Taking 50 Ah or 50% out of the battery and then recharging it, that would be a cycle. Taking all 100 Ah or 100% out of

the battery and then recharging it, this would also be a cycle.

At 100% depth of discharge, the battery is dead. It won't even light up or power anything, it is as empty as it gets. Depending on the inverter, it'll scream and beep at a voltage before absolute 100% depth of discharge. However, there will only be a few percentages left in the battery by the time this is happening.

Most batteries, going to 30% depth of discharge and then back up to 100% does not shorten the life of the battery too much. On most batteries, going to 50% depth of discharge and then back up to 100% is somewhat harmful to the battery. Going to 100% dead depth of discharge is quite harmful to the battery.

You can do this about 12 to 15 times on a car starter battery, you can also do 100% depth of discharge between 150 and 200 times on a marine deep cycle battery. The marine deep cycle battery is meant to deep cycle and it has much better chemistry for doing so.

Every day that your battery is below 100% full, is time off of its life. Every day your

battery is at 100% dead, fully discharged, are many days off of its life.

Depth of discharge is the number of cycles you can get based upon how deep you discharged your battery. As you can see by the graph, taking 100% depth of discharge from full, all the way down to zero on a starting battery from a car, would allow you between 12 and 15 times before that battery was dead.

It won't take a charge nor give up any current. Contrast that to a deep cycle battery doing it for as much as 150 to 200 cycles.

Depth of Discharge (DoD)

The number of cycles you get is based on how DEEP you discharge the battery.

Depth of Discharge	Starter Battery	Deep-cycle Battery
100%	12–15 cycles	150–200 cycles
50%	100–120 cycles	400–500 cycles
30%	130–150 cycles	1,000 and more cycles

So really try to only discharge your batteries to 30% DOD (70% full)

If you have to... go to 50% DoD.

Golf Cart and AGM Batteries are even better than these numbers.

Now, here's where you need to learn some little tricks.

If you go from 100% down to only 50% on a starter battery, you're going to get between 100 and 120 cycles of life out of it, 10 times the life of a starting battery that you might happen to find in a car after a hurricane.

And, if you did only a 50% depth of discharge on a marine battery, you go from 150-200 cycles, to 400-500 cycles. And, if you only did a slight discharge to 30% in depth of discharge and then back up to 100, you're going to get 1000 or more cycles out of a deep cycle marine-type lead acid battery. Now, that's pretty good.

So, really, try to only discharge your batteries to 30% depth of discharge, 70% full and then go back up. You can take it all the way to dead 150-200 times. That's a lot of times, that's many disasters or a very, very long one. If you have to hurt your battery, then you have to hurt your battery. That's what you have it for: Backup emergency power.

If you find car batteries, as in you can find them from vehicles that people are no longer using or are going to be using, like in an extreme crisis, you can use them. But, remember the limitations of them.

The fact is, energy is waiting for you to pick it up many times in a disaster. Let's look at this picture: Free batteries waiting for you, 1, 2, 3, 4 vehicles, all tossed around by a Tornado, all of them with batteries in there just waiting for you to get them and charge up your cell phones and to run your flashlights and everything else just waiting for you.

To recap, there are 3 main types of lead acid batteries. There's the flooded lead-acid battery which is a regular battery. There's a seal lead acid or the SLA gel cell. And then there is the absorbed glass mat, the AGM battery.

FLOODED LEAD ACID BATTERIES

It is time for us to talk further about flooded batteries since this is what a lot of people will end up using. As noted, these are…

1. Car Starting Battery
2. Marine Battery, also called a Deep Cycle Battery
3. Golf Cart Batteries

If you have a serious concern about acid safety, like with children, then you'll be buying the slightly more expensive AGM batteries.

Remember, car batteries are made to provide a maximum amount of current to turn the starting motor on a car, especially when it's very cold and all the oil is thick and the engine is frozen.

These batteries usually never go down by more than a few percent of their full capacity by starting a motor. So, if it's 100% full sitting there, it might go down to 98% starting the car and then get recharged back up by the alternator.

A regular automotive starting battery would not be what you would purchase for a

battery backup system, however, we'll discuss the usage of them because you might end up scrounging or sourcing one of these batteries from abandoned equipment or vehicles.

It'll work as a backup battery and is a lot better than nothing, but it is nowhere near as good as a marine battery or a golf cart battery. They will not have a reserve capacity or Ah rating on them, the starting batteries are only rated in cold cranking amps, which we cannot use to convert over to their total energy capacity ability.

The second type of flooded lead-acid batteries are deep cycle or marine batteries. These have a slightly different lead acid chemistry than a normal starting battery, this allows the marine battery to be discharged deeper, provide energy for a longer period of time than a starting battery and are the same shape and size as starting batteries, but will be labeled as a marine battery or as a deep cycle battery.

For instance, my 2001 Dodge diesel pickup truck uses two 12-volt batteries for starting the engine. They're in parallel, so it is a 12-volt system.

Five years ago, when I had to replace my starting batteries, I replaced them with deep-cycle marine batteries because I wanted to have a better life in the batteries if I used my truck as a power source.

When it's very cold outside, well below freezing, diesel's are known to be hard to start and that is why they have two large batteries to provide as many cold cranking amps as possible. Only when it was near zero at 1am in the morning, did I notice the batteries having any difficulty in starting the truck engine.

So, on cold nights, I plugged my truck in. Most diesel's have an electric cord you can

plug into the wall to keep the oil in the engine block warm, which helps with starting a cold engine.

After I did that, I had no more starting problems so you can use a marine battery in your vehicle instead of a starting battery. I have proved it, you'll lose some cold cranking amps and this might be a problem if you live in Minnesota, North Dakota or Montana and especially Alaska, but not in warmer areas of the world.

Type number 3 of flooded lead-acid batteries are golf cart batteries. Golf cart batteries are a whole other world of lead-acid batteries. Instead of sponge electrodes, they are thicker and grooved.

The chemistry of the battery is much better and there's room under the plates for

when the plates start to deteriorate and collect debris down there without shorting the plates altogether. And there's more room above the plates so you can have a larger volume of water and electrolytes up there.

Golf cart batteries are designed to take a pounding, to be beat up and bounced around on a golf course by idiots who don't know how to drive and don't care. They are designed for fast starts and sudden stops, lots of bumps and a lot of heavy use, which means a lot of depth of discharge.

One thing to note on golf cart batteries is that it is very easy to open the top of the cells and to check the specific gravity of the electrolyte in each cell with a hydrometer. Not even a screwdriver is needed.

Just twist that top right there where the arrows are and all three cells will open up. This is what makes them unfriendly for children. It's so easy to open, even a child can do it. The most common type of golf cart batteries will be 6 volts each.

Two of them have to be put together to make up 12 volts. With these batteries, you can make up very large battery power systems with 2, 4, 6, 8, 20 batteries or more and have a large 12-volt battery system for plenty of backup power.

The golf cart batteries give you the most Ah at the lowest price with the highest durability and reliability. The problem is, each battery weighs 62 pounds apiece and you need two of them. It'll certainly be difficult to pick up your battery system and put it in your vehicle and leave, if you had to.

Once you put it in your house, it's pretty much there. These batteries are what I use for my home battery system, my mother's home system and the ones a lot of my friends and customers use.

There are three common types of golf cart batteries.

1. GC2 - 6 volts apiece (GC2 because you need two of them)
2. GC8 - 8 volts apiece (You have no use for these)
3. GC12 - 12 (You might be using one of these, but probably not)

Look at the picture and you can see my arrows and circles, each one pointing to an individual 2-volt cell.

That's what a lead-acid battery is, it is a collection of 2-volt batteries that are joined internally. Each cell has its own battery with its own lead plates and its own electrolyte, electrically connected to the next cell.

A 12-volt battery has 6 cells, the 6-volt battery has 3 cells. So, continuing the example, a GC8 has four 2-volt cells in it, 4 times 2 is 8, 8-volt GC8. You do not want these, they are not for 12-volt battery backup power at all.

As you can see with the GC12, the large 12-volt golf cart battery, it has six 2-volt cells. I have an arrow pointing to each one of them.

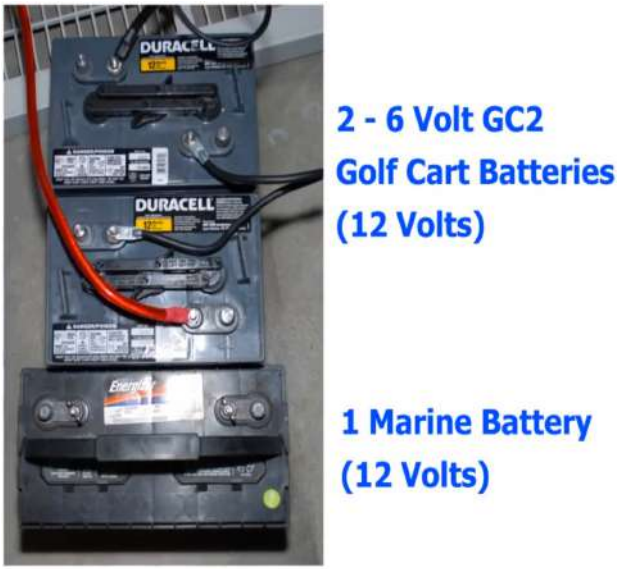
So, with flooded lead-acid batteries, golf cart batteries, you are going to use either:

- 6-volt GC2 golf cart batteries (You can have 2, 4, 6, 8, 10 or more of these)
- 12-volt GC12 golf cart batteries (You can have 1, 2, 3, 4 or more of these)
- 12-volt deep-cycle marine battery (You can have 1, 2, 3, 4 or more of these)
- 12-volt AGM battery (You can have 1, 2, 3, 4 or more of these)

I will show you how to hook these up in the home battery section of this book.

BATTERY SIZE

So, how big is my battery? We have two 6-volt GC2 golf cart batteries for 12 volts on top and we have one marine battery at 12 volts sitting on the bottom.



Let's zoom in closer to these.

To begin, we are going to discuss reserve capacity measured in minutes and Ah. The golf cart batteries on top have a reserve capacity of 120 minutes. The Energizer marine battery, the smaller battery below the golf cart batteries, says it has 210 reserve minutes.

That's much less than 210 minutes for the marine battery, but it says it's done at a 75-amp rating, so it's 120 minutes for these two big batteries and only 210 minutes for this one battery. Let's look at why…

Reserve capacity is supposed to be based on the number of minutes a ‘Trolling Motor’ drawing 25 amps can run for. So, that one marine battery is supposed to run for 210 minutes about three and a half hours on its own.

The golf cart batteries have an RC number of only 120 minutes, but at a 75-ampere rate draw, not 25. If the amperage for RC, reserve capacity, is not specified, it is assumed to be 25 amps. However, I've seen batteries that said 23 amps, so it varies from manufacturer to manufacturer.

It's a numbers game, which is why I always want the true and pure ampere number. We will use really simple math to find the number of Ah of your battery if all you have is the reserve capacity in minutes. All you do is take the minutes, multiplied by the amperes and you always divide it by 60. So, in the case of our marine batteries, 120 minutes times 75 amps, divided by 60, equals 150 Ah.

$$\frac{\text{Minutes x Amps}}{60} = \text{Ah} = \frac{\text{120 min x 75 amps}}{60} = \text{150 Ah}$$

$$\frac{\text{Minutes x Amps}}{60} = \text{Ah} = \frac{\text{210 min x 25 amps}}{60} = \text{87.5 Ah}$$

On the marine battery, using minutes times amps, divided by 60, would be 210 minutes times 25 amps, which is the standard number you will use if it's not mentioned, divided by 60, equals 87.5 Ah.

So, the golf cart batteries are at 150 Ah and the single marine battery is at 87.5 Ah.

150 Ah

87.5 Ah

Remember the GC12 golf cart battery: 77 minutes times 75 amps, divided by 60, equals 96 Ah. It's also nearly $200 for 96 Ah for that one battery, where two GC2s are about $220 for 150 Ah, which includes a core charge.

12-volt marine batteries are smaller and can be moved a lot easier, but golf cart batteries have better lead, better chemistry and better durability.

CHARGERS

There are two types of chargers for lead acid batteries. The main types of chargers are the old dumb type and the new computer-controlled smart chargers that are also called three-stage chargers.

The charger you want is the smart charger. With the smart charger we have three stages in smart charging.

1. Bulk Charging: Putting in the first 0-80% of the battery charge.
2. Absorption Charging (also called the Topping Charge). This is for between 80-95% of the charge of the battery.
3. Float charging. The last 95-100% charge of the battery.

Now follow along with me on the graphic. For the first stage of the charging, the bulk charging, we're throwing in 80% of the energy, that's about 30 amps for 4 hours. So, 30 times 4, that's about 120 Ah thrown into the battery.

For the second stage, from 80 to 95%, it’s about 5 amps over a 6-hour period, that's about 30 Ah put into the battery. The last stage is less than 2 amps for 2 hours, so 2 x 2 is 4 hours. This gets you a battery completely charged in about 12 hours and we took about 169 Ah to do it.

169 Ah times 70% charged efficiency is about 125 Ah, which means that other 45 Ah went to lost charge and to heat.

There's a little lesson I want you to take away from this. This battery got 80% of its energy in one third of time for a full charge. A full charge took 12 hours and we got 80% of it in four hours.

So, if your battery is pretty far down, go outside to start up your generator, have the 30-amp charger going and know that within

the first four hours of charging, it's throwing current in there as fast as it can and you'll get up to about 80% of full charge fairly quickly.

It will take less hours on a 30-amp charger, of course, if you're using a 75 Ah marine battery. It'll do the job quicker if you have 300 Ah from four golf cart batteries.

If you are considering a trickle charger, I would not suggest it. I am not a fan of them. Even if you put a 1-amp trickle charger on there, that'll be 170 divided by 1, meaning 170 hours to charge up. That's seven days of charging, and also, a trickle charger cannot bring a battery back all the way from zero.

1 .(0-80%)	30 Amps - 4 Hours	30 x 4 = 120 Ah
2. (80-95%)	30 -> 5 Amps - 6 Hours	5 x 6 = 30 Ah
3. (95-100%)	< 2 Amps - 2 Hours	2 x 2 = 4 Ah
	1 Amp TRICKLE CHARGER	---
	170 / 1 = 170 Hours = 7 Days	12 Hours 169 Ah

If you go down to 100% depth of discharge, generally, a trickle charger will not charge it up no matter what. Stay away from trickle chargers, put a real computer-

controlled smart intelligent battery charger on your batteries.

You don't need to have the more expensive 30-amp charger either. (If you do want one, I currently recommend this: Schumacher SC1281 6/12V Fully Automatic Battery Charger and 30/100A Engine Starter with Advanced Diagnostic Testing.) But, you can buy a 10-amp charger for $40. The 6-amp version of it is $30.

What is it going to do during bulk charging? It is going to throw in 10 amps for about 12 hours. During stage 2, it'll go from 10 amps to 5 amps for about 6 hours and then during the float charging, it's going to do 2 amps for about 2 hours.

The 10-amp charger will work fine if all you're going to do is charge up the battery before a disaster and then use the battery during a disaster.

TESTING YOUR BATTERY

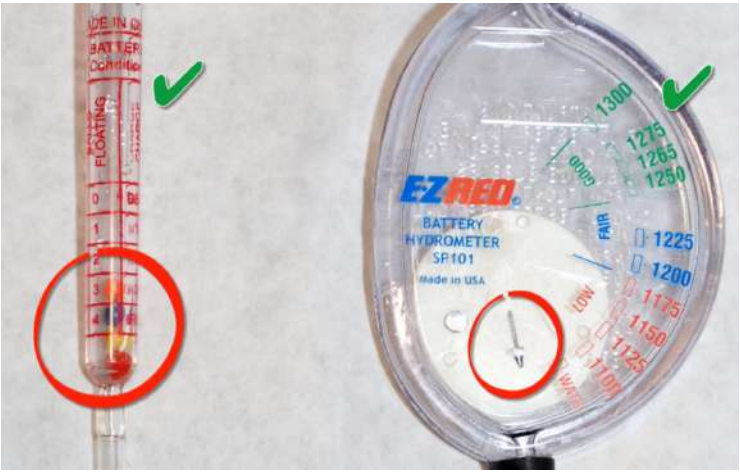
The quickest, the easiest and the most reliable method there is, to know if your battery is good, is to test the sulfuric acid. Caution! You're going to have to wear goggles

and rubber gloves. You are going to be working in and around sulfuric acid.

When testing the battery and acid safety and measuring the specific gravity of the electrolyte (the electrolyte is sulfuric acid and distilled water), you're going to use a hydrometer. Basically, a hydrometer tells you how dense the fluid is.

You use the little squeeze bulb at the top of the hydrometer and you suck the acid up into the meter with the squeeze bulb.

You'll see that the acid hydrometer type on the left has four floating balls in it, they are all the way down at the bottom. When the sulfuric acid is at the right density and the battery is completely charged, the four balls will be floating up near the check mark.



For the easy read meter on the right-hand side up by the check mark, you'll see the green numbers. You'll want to see that dial rotate and be up there.

I’m now going to show you how to test the acid in our batteries, starting from the top and going counterclockwise.

Notice, we have the golf cart batteries labeled at the top and then we have the marine batteries on the left and on the bottom left I am pointing out the battery caps on the top of the marine batteries with the arrows.

Then we have the hydrometers, the safety glasses, the screwdriver for popping the caps on the marine battery and we have thick rubber kitchen gloves.

We also have a small pail of water to rinse off any sulfuric acid that might get on our fingers or our tools. Of course, we have some paper towels and up there near the top, I am pointing to the battery caps on the golf cart batteries.

We are going to test the acid inside of each cell to make sure that all of the balls are floating and you have the highest specific gravity that you possibly can. When you have the highest specific gravity, it means that the batteries are charged all the way up.

Now, look at that picture again, what did I tell you a 12-volt battery was? A 12-volt battery is really six 2-volt batteries inside of one case. So, in the picture above, I clearly have it marked with the lines.

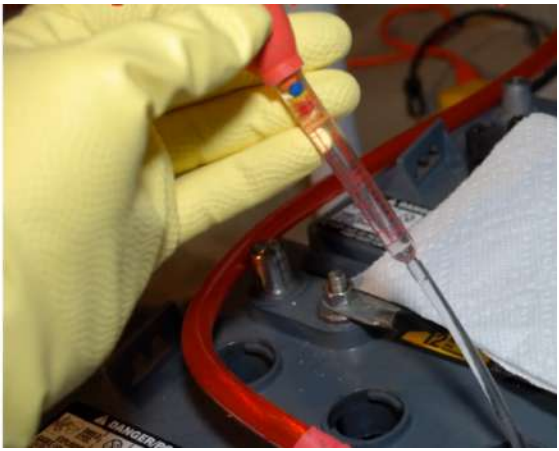
The lines would be where the dividers are separating the batteries. Each one is its own battery. So, if one of the cells doesn't test right, then the whole battery will not be working correctly because the battery charger will never be able to get the whole thing up to charge.

When testing the acid in the golf cart batteries, put on your safety glasses and your kitchen rubber gloves. Put your finger and forefinger on the levers on the top of the battery caps and twist them, then lift up.

You will see that the three caps are all connected to each other and you are now looking down into three separate holes, into three separate 2 volt batteries, each one with its own sulfuric acid.

The sulfuric acid is the clear, water-looking liquid that is in the battery. It is not water, it is sulfuric acid and water, be careful!

You are now going to take the battery hydrometer, squeeze the red bulb, then put it into the battery acid.



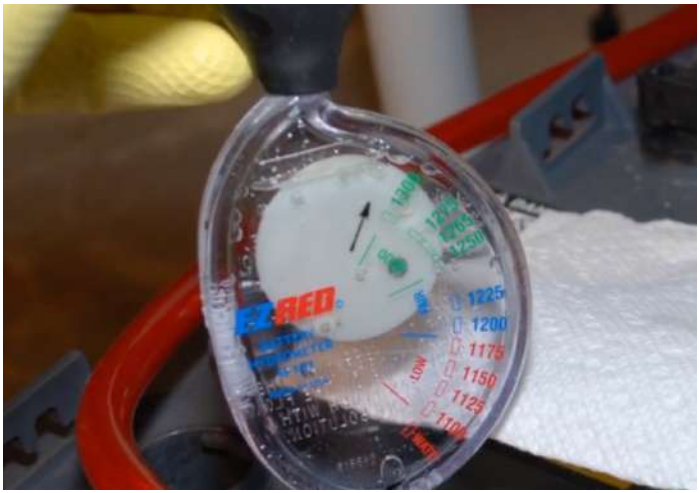
You’re going to let go of the red bulb and it is going to suck up a volume of acid into the hydrometer. Look up near my fingers and you'll see the four balls sitting there.

These four balls are all the way up at the top, which means the battery is good and has a very dense liquid in it. If the battery had a lower density of sulfuric acid in it, it was not

fully charged or had a problem with the cell, one or all of the balls would be down towards the bottom.

If you have three balls up and one ball down, you know there is something going on with the battery, it's not fully charged. Charge it some more and then retest the sulfuric acid.

We're going to try the other sulfuric acid hydrometer to measure the specific gravity of the acid and the water. As you can see, it is pointing all the way at the top above the green.



However, I have come to learn I don't like this one very much. You have to squeeze that bulb all the way down until it's flat and then suck up a whole bunch of sulfuric acid. If

you don't suck up enough to fill the whole thing, you are only going to get a partial reading, leading you to believe you have a bad battery when you don't.

So, I would avoid this type of hydrometer and stick with the 4-ball type of hydrometer that you can get on Amazon or at Walmart.

As I said, the battery is made up of individual 2-volt cells. We just tested the one all the way to the bottom right, now we need to do the same thing on the middle cell or cell number 2 in this 6-volt golf cart battery.

And then we're going to move and do the same thing to cell number 3 of this battery.

Be careful! Notice, in the picture below that I dribbled a little bit of sulfuric acid off of the hydrometer when I was moving it out of the battery acid. I dripped some acid onto the top of the battery. We are now going to clean this up safely.

You are going to take a brand-new paper towel and you're going to put it into the bucket of water with your gloved rubber hand

and you are going to get it halfway soaked with water.

Next, squeeze out some of the water so the paper towel is nice and moist. Wipe up your little acid dribbles that you dribbled onto the battery with the moist towel. I will show you how to safely dispose of the moist towel with a little bit of battery acid on it shortly.

Now, take the caps for the golf cart batteries and put them back on so everything goes into three holes. Then turn the lever in the opposite direction and lock them on nice and snug.

We are now going to test the sulfuric acid in a standard marine deep-cycle battery. For this, you are going to need a screwdriver. There will be a little indentation in the cap on the battery.

There are usually two caps, one on the left, one on the right. One of the three cells on the left, one of the three cells on the right. So, insert the screwdriver into the little depression, press down and lift out the cap carefully.

As you can see, the cap just lifts right off and it covers the three holes. There can be a little dirt and debris down there, which we will clean up shortly. But first, we need to test the density of the sulfuric acid in cell number 1, then we need to do it in cell number 2 and 3.

Next, you will need to test cells 4, 5 and 6, the exact same as we have done in cells 1, 2 and 3 of the golf cart battery and of this marine battery.

Now I'm dipping the hydrometer into the little bucket of fresh water just to get any acid off the bottom of the tube. I'm not sucking water up into the hydrometer, I'm just rinsing off the outside of it, so when I put it down, if I

touch it with my hands, I'm not touching sulfuric acid.

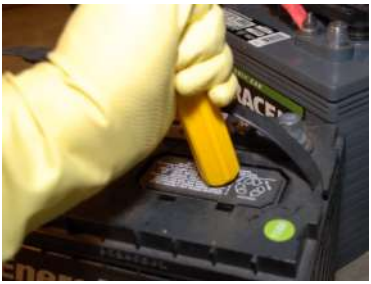
Back to the battery, let's look at cleaning up the holes on the top.

Notice I am cleaning up the top of the battery and am going in a circular fashion around the cell, being very careful not to push any of the gunk into the battery, but to make sure it sticks to the moist paper towel.

You know those plastic shopping bags that everyone hates, well they are really good for disposing of rags with sulfuric acid on them. Have a shopping bag near you and dispose of your paper towels into the plastic bag. Once you're all done, tie it up by the handles and throw it into your regular garbage.

Let’s take the cover to our battery acid, and snap it back on to the top of the battery. You might have to pound on it with your fists to make it snap all the way back in, but you'll get a reassuring snap when it does go in. If it

doesn't go on easy enough with your fists, just use a screwdriver or a hammer to tap it back in there gently.

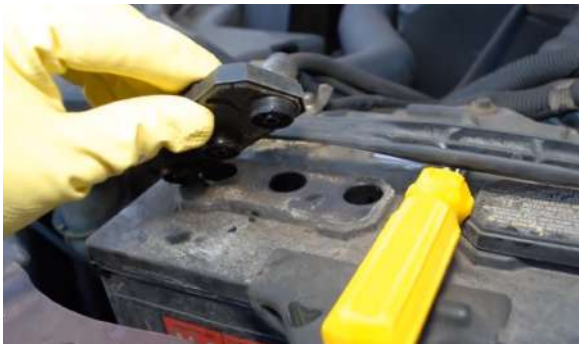


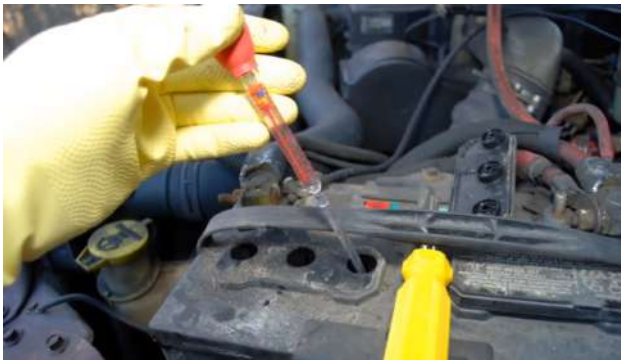
Go and do the same thing on the cells on the left that you did to the cells on the right. It is exactly the same procedure.

Alright, let's go check the acid in my 2001 Dodge diesel pickup truck. The batteries in there are about five years old and I haven't done a thing to them.

It’s the same thing with these batteries as it was the batteries inside. Put the screwdriver underneath the cap and lift and then using your rubber gloved hand, remove the cap from the battery and put it in a safe place. Put the hydrometer into the battery acid and suck it up.

As you might be able to tell in the pictures below, all four balls are floating all the way at the top. I have a fully charged and good battery cell on this cell of the battery.





Let's move to the second cell. I can't pull any acid out. I've put the hydrometer in at the top of the plates and it doesn't suck up anything. My acid is below the plates and I have low water level in my battery.

Looking down into the battery with a flashlight carefully, I can see that one of my cells has liquid on it. All the way to the right, it has water and sulfuric acid in it and the other two aren't all the way to the top.

It's somewhat below the level of the plates, it is not dry. The battery is working, but it's just a little bit below the plates. So, we are going to need to add distilled water to them, not deionized, not mineral water, not spring

water, not the water from the creek… pure distilled water you buy in a gallon jug at the grocery store.

You’re going to have to use a funnel or be really good and pour the distilled water directly into the cell.

Do not overfill it, because when you do, you will have not only water, but also sulfuric acid spilling all over your battery.

If you're being careful like I am, you look at the arrow on the picture and see there's these little plastic fingers sticking down into the lead acid battery. You want to fill the battery with distilled water until the water is to the very bottom of the plastic finger sticking down the hole.

If it's a little bit above the bottom, fine, but that is the level where you want it. Not halfway up the fingers and not at the top, because the battery will heat up, everything will expand and the water and acid will be squeezed out. Only fill to the bottom of the fingers.

In case you couldn't see it in the previous photograph, here is a real close-up photograph of what I'm talking about of where the acid needs to go, to the bottom of the plastic fingers.

After you are all done and before you take your gloves off, be sure to wash the exterior of your gloves either by dipping them into the water bucket full of water or running water from the garden hose over them.

You want to be very careful about dripping sulfuric acid onto the floor that you're working on for several reasons. One, you do not want to leave any of the acid sitting there in the form of droplets. Wipe it up with a moist towelette.

And I guarantee you, if there's any sulfuric acid on the floor, whether you have a pet or children running around, they will find it and it will get on their hands or their paws followed by irritation on the skin.

If for some reason you start developing this nice gentle warm sensation on your hands or your skin, your arms or your feet, it means you have sulfuric acid on that part of your body. Please, immediately go to a water source and rinse off this part of your body with copious amounts of water to get rid of any sulfuric acids.

Remember, when testing the battery and acid safety, each cell is its own two-volt battery. If one of the cells has a bad acid reading, then that entire battery will not charge correctly, which is not really fixable.

This is not common, but it does happen which is why we test the acid in all of our battery cells, especially, if they aren't charging properly.

Regarding hydrogen: It is normal for a very small amount of hydrogen and oxygen to be produced during the charging of a battery. Hydrogen is 1/14 the density of air, it does not linger around.

Any amount of hydrogen made is leaving your room, your house, the entire building at a rate faster than a human can run. It is trying to instantly dilute itself across the entire planet.

You do not need to ventilate your space. There will not be any hydrogen build-up anywhere. There is only one place where hydrogen build-up might possibly be an issue and I will show you where that is…

Above, is the only place where hydrogen might build up, underneath the caps of the batteries.

When you put on jumper cables, battery cables or charging cables, there is the possibility of creating a spark on the battery terminal when you are doing this.

]

Notice I actually took a photograph of sparks being produced on the battery terminal. Hydrogen could build up under the caps of the battery and a spark very close to the caps could ignite the hydrogen.

If there was enough hydrogen under the caps or leaking out at the precise moment of the spark, and if the spark was in the right proximity of the hydrogen, it could ignite.

All of this is unlikely, but it's why I tell you to wear eye protection and rubber gloves when working around lead acid batteries.

So, how do you prevent from making sparks on the battery terminals?

This is a smart computer-controlled three-stage 2012 charger and it's dumb enough that it sends current to the clamps

when it's on and when you put it on the battery, it might spark. So, what do you do to prevent any type of sparks?

That is easy, you put the clamps of the battery charger onto the battery and notice the battery charger is off, it is unplugged from the wall.

Now, take the 120-volt cord coming off the battery charger and plug the battery charger into the wall or an extension cord so the battery charger can have 120 volts.

Now, the clamps are on and the power is going to them and since you're not placing them on with power going to them, they will not spark.

Okay, if you have reached this point in this book, you should have come to a

conclusion about if you want a marine deep-cycle battery. Do you want one or multiple of them? Once you start getting past four batteries, you really should be going with a golf cart battery.

So, your other choice is a golf cart battery? How many do you want? 2 is a great place to start, 4 is also a solid place to start.

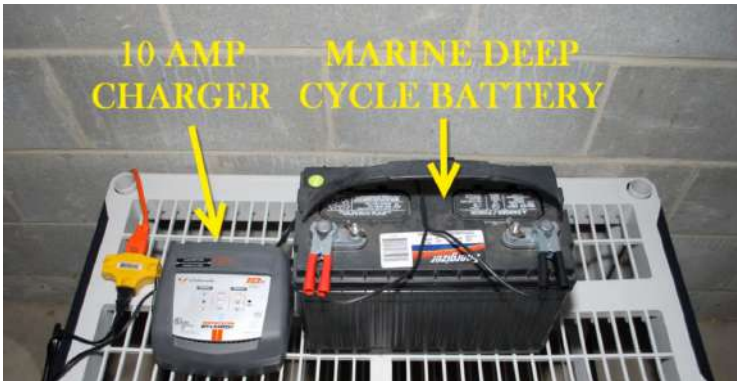
But, once you start getting into 6, 8 or 12 batteries, it becomes a big wiring job, but if that is what you so desire, that is what you should do.

As noted, your third option is if acid safety is a concern for you. If so, then you will be choosing an AGM battery, such as an Optima, Odyssey, Duracell or any other manufacturer of an AGM battery. You will treat these as you would the marine deep-cycle. You can have just one or as many as you desire.

MAKING YOUR BATTERY SYSTEM

For your home battery system, I will give you the simplest version, a marine deep-cycle battery or maybe an AGM battery with a 6 amp or a 10-amp charger on it. Do not go below a 6-amp charger. For instance, you

could buy the Marinco ChargePro 6 for 100% silent charging.



The charger's clamped onto the battery with the red on positive, the black on the negative, with the charger plugged into the wall. The charger was turned on to its fast charging mode.

After it has charged all the way, it will go into a 'maintain mode'. You can leave the battery and the charger hooked up like this forever and you should. It should be continually hooked up until the power fails.

When the power fails, you clamp on to the battery with a cigarette lighter socket that has two clamps on it, one red, one black and then plug it in. In this example with a 4-way DC outlet, I have four outlets for one plug.

You can find them with two, or three, but this is the one I like the best. In the photo, you can see I have two USB phone outlet chargers and a voltmeter plugged into it.

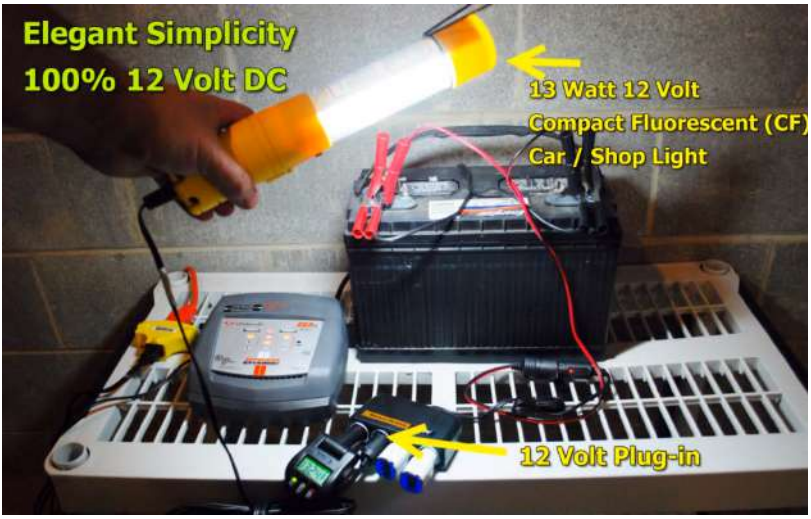
This is really what I call elegant simplicity. This is a very simple 100% 12-volt battery system for the basics.

I have two chargers in there, both of them have 12 volts to USB chargers and both high amp DC chargers for the USB. One socket is 2 amps, the other one is a 1 amp. So, I can charge four USB devices simultaneously.

Plus, I have the voltmeter on here, which is very important to have on the system all the time, because it lets you know the health of the battery. As far as voltmeters, I like the INNOVA 3320 Auto-Ranging Digital Multimeter.

When you are running a basic and simple battery system for emergency backup power, it is basically about two things: Communications, thus recharging your cell phone, your iPad or whatever you want off of

a USB, and it is about light. Being able to run light and being able to talk on the phone.



This is a DC compact fluorescent light, it's 13 watts, has a cigarette lighter plug on it, and plugs right into the socket. It gives you more than enough light for whatever you might want to do and has a very long cord on it.

As you can see, the light just plugs right into one of the open sockets. So now, I have a voltmeter, a light, and 2 USB chargers. If you need more sockets, buy more four ways and just keep plugging them into each other. The four ways I like are the “Wagan 4-Way 12V Automotive Socket Splitter and Extender, Cigarette Power Splitter.”

You can run quite a bit of low wattage things such as more USB chargers or 2-3 more lights, but always have your voltmeter in there so you know what is going on. It is very important to have a system health monitor on something that you are going to depend your health and safety on in the future.

Now, we are starting with the most elegant bare-bones DC system and then I am gradually expanding it and making it bigger for

you. I have now plugged in a simple $20, 150-watt inverter into one of the DC sockets that is converting the 12 volts DC to 120 volts AC. BESTEK is a company that makes solid inverters and you can check them out on Amazon.

As you can see, I am running a five-watt compact fluorescent light bulb in a house lamp.

It's getting a little crowded in the single four-way, but like I said, you can have more than one. You can plug a four-way into a four-way into a four-way… As many times as you desire, as long as you're running low wattage appliances off of it.

I have plugged in the inverter and notice it also has a USB charger built into it, so, two is one, one is none. I have my regular USB chargers and I can also recharge USB off of this inverter. It is a 2.1 AMP and a 1 AMP charger so it will fully charge up an iPad as well as an iPhone or an Android device very quickly.

Now, you're not going to run your refrigerator, freezer, or your sump pump off of this. This is merely for the convenience of being able to run smaller items.

You can plug the biggest extension cord you want into this and run your extension cord all around the house as long as you are putting just compact fluorescent and LED lights on it to illuminate your house in the dark.

Now, let's go up one size bigger. I've taken off the 150-watt AC inverter and all the DC items and have plugged in a 400-watt inverter. (BESTEK makes a good 400W inverter.)

This clamps on to the batteries. Most small inverters, 150 watts or less, have cigarette lighter plugs on them. So, if you have a 400-watt inverter and it has a DC cigarette lighter plug adapter on it, it'll only be able to pull about 150 watts out of a cigarette lighter.

That's limitation, it doesn't matter whose brand, it's limitation of the cigarette lighter, you're not going to pull more than 150 watts out of it.

All the bigger inverters have clamps, the 400-watt, 800-watt, 1200-watt, 1500 watt. I clamp onto the battery, and as you can see in the picture above, this one is clamping on and it is running the same 5-watt compact fluorescent light.

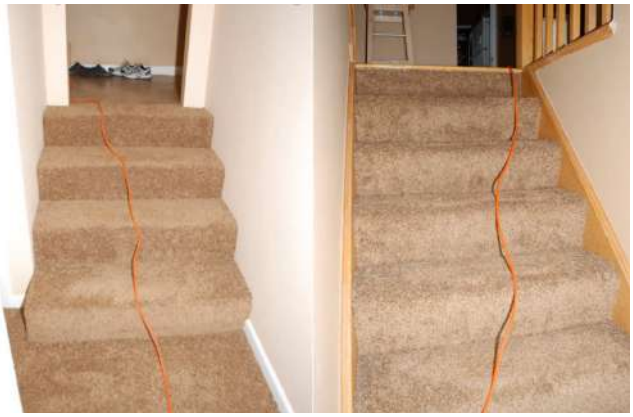
Now, in a disaster, you don't want to move your living area to where the battery is, in this case, the basement. You either want to move the power from where the battery is up to the living area or you want to move the battery because it's only about 50 pounds.

Notice what I've done here, I have plugged an extension cord into the inverter, just a regular orange extension cord. It will handle up to 15 amps, which is a lot more power than the 400-watt inverter can put out.

If you wanted to get a light duty extension cord, such as a 16-gauge extension cord, you can find one that is 120 feet for around $24 and run it all the way around the house as far as you want.

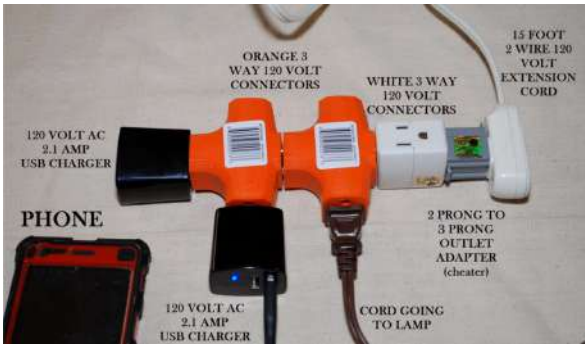
You can run it from the car outside into the house, it doesn't matter. This inverter is only going to put out 400 watts and almost any extension cord will be able to handle that.

I have taken this extension cord wire and have run it all the way up one set of stairs around the corner and up another set of stairs. Then I ran the extension cord to the dining room table.



This is a table that is upstairs in our kitchen, a good 60 feet or more of cable away from the battery in the basement. The extension cord is coming up from the back of the table and I have a compact fluorescent light running off of it.

You can see I also have some adapters there that I want to go over. I'm doing some fun stuff with these and it's very easy, very affordable and it's better than a power strip.



I have a 15-foot, 2 wire, 120-volt white extension cord plugged in to the orange cable that was coming upstairs.

This is a two-outlet extension cord, so I have a two prong, three prong outlet adapter on it. Then plugged into there, I have a white 3-way 120-volt connector, these are available at Walmart, Amazon, you name it, and it has three connections on three sides.

Into that, I have plugged in the famous orange ‘T’ adapter. I love these ‘T’ adapters because I can plug in seven of them into each other and have all the outlets I could possibly want. They're extremely durable, even a car can run over them. They do not have the fragility of something like a power strip does and they work great.

Into here, I have two 120-volt AC 2.1-amp USB chargers, each one with two USB outlets on it. I also have the cord going to the compact fluorescent light that is lighting up the whole room quite nicely.

This is what I suggest you do with your battery system where you are going to move your power and hook everything up to it.

My phone is hooked up to it and charging, my iPad can be hooked up to it, your AA recharger can be hooked up to. There is no end to what you can hook up to this configuration and if you need more outlets, you can just keep on expanding out.

As I mentioned, you can either have the battery in the basement with the extension cord running upstairs or you can bring the battery and the inverter upstairs and set it on your kitchen table.

There's nothing wrong with this, there is nothing harmful about it, just don't drop it, don't let it fall off the kitchen table and hit the dog, hit the floor, or hit someone's toes.

Let's look at going whole hog crazy. This is how you can set up a marine battery on your kitchen table. I put this together for you

and will explain starting at the top and moving clockwise.

I have the marine battery which is a group 29 deep-cycle marine battery. Attached to this is a 400-watt Duracell inverter.

It is plugged into the orange extension cord and is going around the battery and then to my outlet strip made out of ‘T’ shaped adapters. Next to the Duracell inverter, I have another 400-watt inverter, then a police scanner.

I have the lamp from around the house with the compact fluorescent light as well as an iPad being charged off of my USB wall charger where I have three of them plugged in at the same time.

Next, is the plugged in Android phone, my Trent external lithium ion battery, which is enough to recharge my wife's iPhone sitting next to it three times. You will also see an Android tablet being recharged, as well as a two AA headlamp flashlights.

Above that is the AA and AAA battery charger. It will charge 8 AA or 8 AAA batteries in less than two hours.

Above that, I have a simple desk lamp, a radio which runs off a battery, so it runs off of 120 volts. In the middle, below the battery and the radio, I have what you've seen before, the 4-way DC outlet. Into this I have plugged in the voltmeter to give me the health of everything that I'm using.

I have a DC USB charger and a cable plugged into it. There is a 150-watt inverter plugged in just for illustration purposes. I have three inverters here, any of the three could power everything sitting on the table easily without any problems.

If you start out with the 150-watt inverter, you can always upgrade to a 400 or 800-watt inverter later on.

This setup covers most things you can do with it. Just keep it as simple as you can get with one marine battery sitting on your kitchen table.

There were a few additional items missing in the last photograph. I wanted a small portable LCD television that runs off of low power.

The one below runs off of nine watts of power. It is also a DVD player and came with a DC cable, an AC cable and an antenna adapter that you'll need to plug into that RCA rabbit-ear antenna. I have it plugged into the inverter going to the battery.

That battery will run the TV for a long time. This is the type and size of a TV you want to run off of a marine battery, otherwise, you're going to drain it down really quickly.

I cannot emphasize how important it is to have an external antenna even like this little one. I get over double the channels coming in just on this little pair of rabbit ears, then over the antenna that came with the TV.

BATTERY CHARGERS

Before we go further into batteries and go up one size in the system, let's take a moment to talk about battery chargers. Up until now, my favorite battery charger was the Vector battery charger, the one you see on the right. One problem we discovered is that after you leave them on for about four years, the fan eventually dies in them.

Schumacher is a company in the United States making a computer-controlled, intelligent three-stage battery charger.

The one on the left is the 30-amp battery charger, the one in the middle is the 10-amp battery charger. Now, they also sell

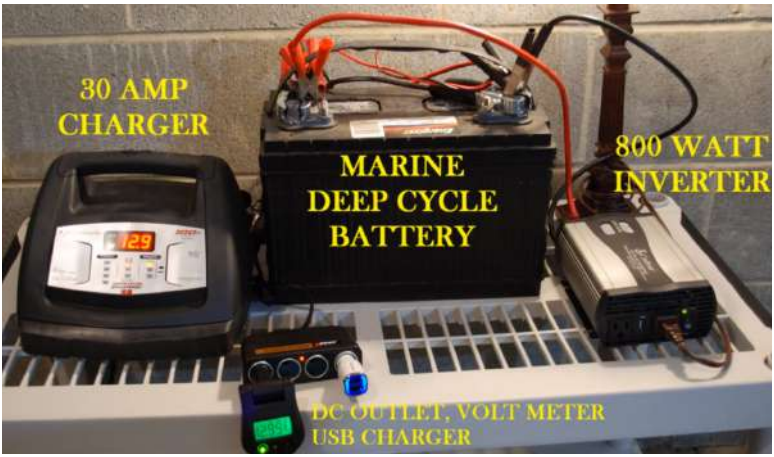
them in 6 amps, but that is as low as you want to go.

You never, ever want a trickle charger, you want a full intelligent computer-controlled charger like these. These have a full ‘Maintain mode’ in them, which will maintain the battery and will bring it up to 100% and then they will maintain it at precisely 100%.

In the time since I’ve created this book or in the future, if Schumacher is no longer on the shelf, what are you looking for? You are looking for a computer-controlled three stage battery charger. That’s it.

Okay, now we're going to go up in size a little bit more. We are keeping the same deep-cycle marine battery that we had previously and I have gone up to a 30-amp Schumacher charger instead of the 10 amp one that you saw, for several reasons.

The Schumacher 10 amp is like $40 and the 30 amp is about $75. If that price difference is not significant to you, buy the 30-amp charger, it has a display on it, it's bigger and it’ll charge your battery quicker.



Again, notice we have our instrumentation in our DC outlets sitting there ready for us. If the inverter were to die and you only have one, you might be stuck with DC, so onto here I have placed an 800-watt inverter.

Why do I have an 800-watt inverter? You don't want to run 800 watts off of the marine battery all the time, it won't last very long. This is not something you're going to plug your refrigerator into. However, you might have a sump pump and running a sump pump when it's filling, overflowing with water, is generally a high priority for people.

A sump pump does draw a lot of power, but it only does it for about 30 seconds at a time. So, with this marine battery and inverter,

it depends upon how much water is flowing into the sump pump and how much it's raining.

This might buy you a day or two of sump pump backup, which is a lot better than bailing water out of a sump at 3am and carrying it up the stairs to throw it out.

So, this is one step up and you are going to see us expand on this even further. While we're on the subject of the charger and the larger battery banks, let's take a moment to review the charger and the charger's operation.

No matter if you are on a 6-amp, a 10-amp or the 30-amp charger, when you see the green light, it means that the battery is all the way charged. On the bigger chargers, it'll say 100, meaning 100% fully charged.

The button on the left is what decides how fast you are charging. Push the button once and it lights up the snowflake.

This snowflake is in ‘maintain mode’. It’ll run between 1-2 amps to a fraction of an amp, ever so precisely into the battery at the right times. Only when it needs it to in order to precisely maintain the battery at 100% state of charge.

It also has a built-in desulfation routine, which will run periodically to make sure the battery is not sulfated, which is what you want in an intelligent computer-controlled charger.

If you push the button again, it will move down to the turtle, which will charge your battery at about 3 amps. This is not what you want to do.

When you push the button again, it will move down to the rabbit and this means it will charge your battery at a rate of up to 30 amps.

This is what you want, when you put the battery charger on the battery and plug it in the wall for the first time, you want to put it on the rabbit.

Also, if you are running a generator outside and you wish to recharge your marine battery, you're going to want to do it as fast as possible, so you will want to put it on the rabbit symbol.

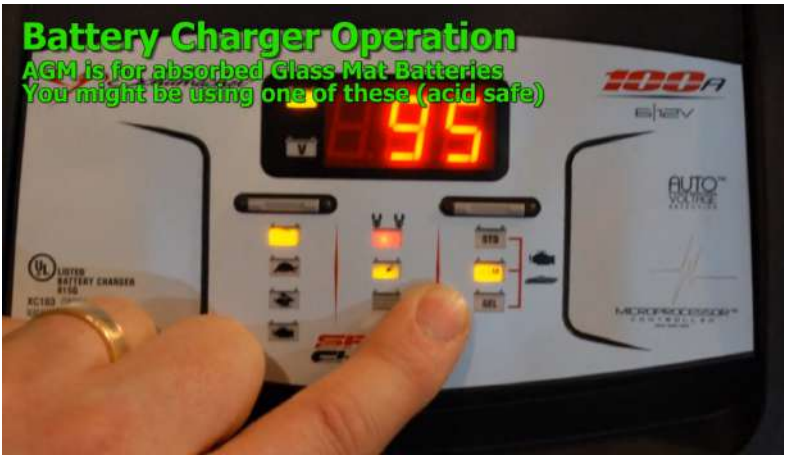
The last symbol is a little engine and this is for the little engine that can't. We do not want to run this on a little engine symbol, it is for giving your car a little bit of a boost when you're trying to start a dead battery. This is not a feature we are going to use in any shape or form on our home battery system.

The button on the right tells the charger what type of battery you are on, because depending upon the type of battery, it determines several things: Its maximum voltage and its rate of charge and how it desulfates.

So, if you push the button once on the right-hand side, it is going to light up the light that says STD on it, for standard. This is your standard lead acid marine deep-cycle golf cart battery, your starting battery.

Also note that it says 95 on the display, that means the battery is 95% charged. It will work its way up to a hundred and then the charger will go into ‘Maintain mode’.

If you push the button again, it moves it down into a ‘AGM mode’. For those buying the lead safe batteries, the AGM batteries, the absorbed glass mat batteries, you want to make sure the charger is in AGM mode for charging the AGM battery.



The third mode of charging is the least aggressive mode of charging and it is for gel cells. Let's say you have some gel cells out of some computer UPS's or salvage dump.

I don't really want you using salvaged batteries because you don't want to trust your health and life and safety to a questionable battery. But, if for some reason, you have gel cells available to you and you're going to be charging gel cells with this charger, make sure you push the button and move it down into the gel mode.

When the time comes that you need your battery bank, you might be frantically thinking “Well, when did I want, the rabbit, the turtle or the engine?” So, I write cheat sheets.

I take blue 3M painters tape and write down helpful information to get my system going without a hitch.

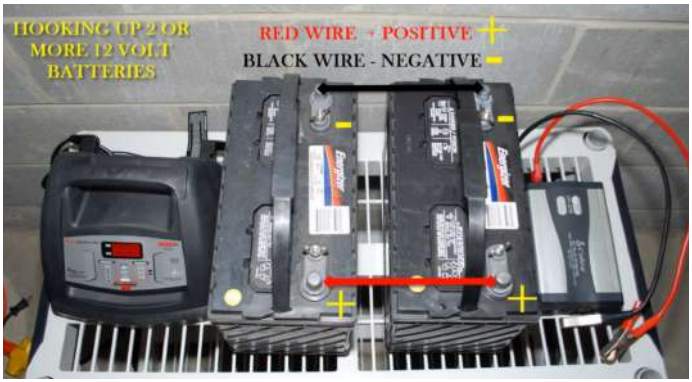
It works and will save you. Put your own cheat sheets on your devices so that when it comes time to repeat it under stress, you can do it at a glance or someone else in your family who's not used to doing it can push the button that needs to be pushed.

I like very much to have a light on all the time on near my battery system. In this case, I use a little 5-watt compact fluorescent light, which is all you need to have on or an LED light.

This does two things: One, it lets you know that the battery system is working, just by you glancing at it. Two, if you see that the light is not on, you know you have a problem.

So, what if you want an even larger battery bank? We can do that. The first step we will take in making a larger battery bank is by having two or more marine deep cycle or AGM batteries connected to each other.

This can be done by hooking up positive to positive, plus to plus and negative to negative, which is minus to minus.



It is very important you get this right. If you can't tell which one's positive and which one’s negative because they don't have any colors on them, you can always find the mark embedded in the plastic by the terminal.

Instead of showing you how to make your own very thick cables for connecting batteries together using a 2-gauge wire, we will choose to start with simplicity.

In this example, I just picked up two battery cables from Walmart, they can also be found on Amazon.

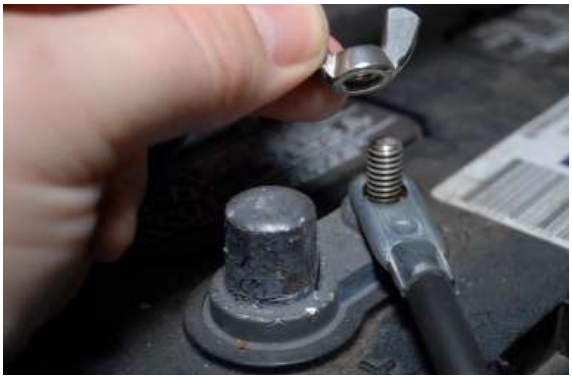
I purchased the shortest ones possible, which at the time were 19 inches long. You definitely want to go with jumper cables that are 4-gauge or smaller.

We are going to take one of the 4-gauge cables and are going to jumper it across the two terminals in the front, these are the positive terminals. Notice how the batteries are lined up in the picture.

Both terminals are on the right side of the batteries.

Our jumper cables have eyelets on each end, so all you do on your marine battery or you AGM battery is unscrew the screw off the terminal, put on the wire and then you screw it on finger tight, just with the wing nut.

If you don't have a lot of strength in your fingers for some reason, then I would advise you get a wrench and tighten it all the way down, but normal human finger strength will be fine.



Now I have put both cables onto the battery, going from plus to plus and minus to minus. I have the charger on the left, battery 1 and battery 2 hooked up in parallel and the inverter on the right-hand side.

Notice right now the charger or the battery is not connected, we are about to do that.

In order to hook up the inverter to the battery, you are first going to take the red clamp from the inverter and hook it up to the positive terminal. Then you are going to take the black clamp from the inverter and hook it up to the negative terminal.

Now we're going to hook up the charger. Notice the charger is unplugged and not lit up. The Schumacher charger likes to put voltage to the cables even when it's on, even though it's not charging, so when you put it on the battery, it'll spark.

So, with the charger off, I'm putting the red cable on the positive terminal and the black cable on the negative terminal.

Then we plug the charger into the wall so it has 120 volts going to the batteries. Push the little button on the left so it goes down to the rabbit so it'll charge up to 100% at its quickest rate.

Next, we are going to charge everything up and get ready for our emergency.

Here are some basic rules for you…

- You can put together anywhere from 2 to 20 marine or AGM batteries. These batteries should be all the same age and should be bought at the same time, otherwise, one bad battery will bring the whole bank down.

- All your batteries will be going from plus to plus and minus to minus. Don’t mix 125 Ah batteries with 75 Ah batteries, you're not mixing group of 31 in group 29s and group 27s with each other.

- Keep your batteries inside, they should be cool in the summer and warm in the winter.

- Battery 1 and 2 connected together gives you more life than using up all of battery 1 and then switching into all of battery 2.

- Battery systems are for small things such as phones, small TVs, radios, AA chargers, iPads, DVD players, etc. Powering refrigerators and freezers are not for small battery systems. These are best powered by an inverter and your car and/or a generator if you want. If you wanted to power your refrigerator freezer for a short period of time, you would have to have a very large battery bank and I mean 400 to 800Ah or more of batteries.

Now, I keep on telling you to connect positive to positive and negative to negative, so let me explain why.

Notice the terminals on one battery are on the right side and the terminals on the

other battery are on the left side, showing positive to negative and negative to positive.

If you did 12 volts plus to minus and 12 volts plus to minus on those two batteries, you would instantly short-circuit everything.

It would melt the cables and cause an instant fire. It would catch the plastic of the battery boxes on fire and then vaporize sulfuric acid into the air and you would get a full-blown structure fire. That is why I tell you not to mix it up. When you are working with 12-volt batteries, it will always be plus to plus and minus to minus.

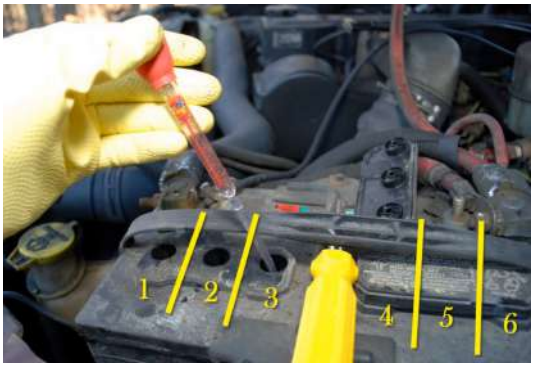
If two, three or four marine batteries isn't enough for you and you want a larger battery bank, we will cover that now. As I've said, golf

cart batteries are really two 6-volt batteries. It is no different than what we have been using.

I mentioned that a 12-volt battery was really six 2-volt cells inside this 12-volt battery case, they are batteries on their own. They each have their own sulfuric acid and are connected internally electrically for your convenience so you don't have to hook them up on the outside.

Here is a regular marine battery. You will see that it is, like the 12-volt golf cart battery, divided up into six cells.

To further illustrate it, here is the battery with one of its caps off when I was checking the specific gravity with the hydrometer.



You can see that cells 1, 2 and 3 along with cells 4, 5, and 6, make 12 volts.

It just happens to be that instead of having all 6 cells in one case, we have three cells in one case and three cells in the other case.

Let's connect these two batteries in series, which means the plus on one battery goes to the negative of the other, six plus six will equal 12.

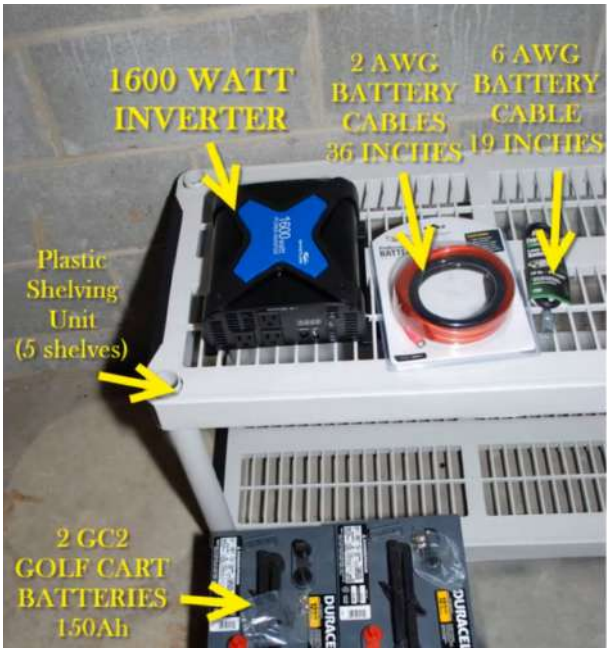
Look at the green arrow in the picture. This is where it is going to be jumpered with a cable and then, once we jumper it, what do we have?

We have a 12-volt battery. You will treat this no differently after it is jumpered, than a 12-volt battery

Once you have this set up as a 12-volt battery and you want to add more golf cart batteries, what are you going to do with your 12-volt battery? You are going to go from plus to plus to plus to plus to plus and minus to minus to minus to minus.

So, we are going to start with 2 golf cart batteries or GC2. I have a plastic shelving unit just to make things convenient and I have a 1600-watt Whistler inverter sitting there which will be on the golf cart batteries.

Now remember, 1600 watts doesn't mean you can run it at 1600 watts all the time, it means the inverter can handle up to 1600 watts.



I have awfully thick cables. When you are dealing with a 1600-watt inverter or higher, you need to have thick cables, 1000-1500-watt inverters will take 2-gauge, American Wire Gauge cables.

They shouldn't be longer than 36 inches and they do come in red and black. I also have a six AWG patch cable that is going to go between these two batteries to make the two sixes into one 12-volt battery.

So, the first thing we are going to do is to take the plastic caps off of the battery

connectors and throw them away, we do not need them.

Now, look at this next picture very carefully.

I have the plus of one battery on the left and I've circled where it says plus in the plastic. I have the negative of the battery on

the left and I've marked it as negative and I have circled where it says negative in the plastic.

On the second battery, the one on the right, I have marked where it is positive and I have circled in red where it is plus embossed in the plastic and the same thing goes for the negative on this battery.

My fingers are going across the negative and the positive of the two batteries we are going to connect.

You will connect together two 6-volt batteries to make one 12-volt battery and again, in this example, I am going to use one of the 19-inch 6-gauge wires that I got for connecting the batteries together.

Notice I have put the cable eyelets over the battery post. These golf cart batteries do not have wing nuts on them as many of the deep-cycle marine batteries do, they have regular nuts on them that you are going to have to tighten down to hold the cable in place.

Screw down the nuts over the eyelet of the cable whereas with the wing nuts and the smaller marine batteries, you might have been able to tighten them good enough with your fingers to make a good connection.

But, with these, you are going to have to use a wrench to tighten these down as tightly and as snug as you can.

You now have a 12-volt battery! The positive post on the left is where the red wire will go to the inverter. On the right-hand side, the negative, the black wire, is going to go to the inverter. When you buy an inverter, don't forget to buy cables. The cables don't come with a lot of inverters anymore.

Okay, let's do some work on the inverter. I personally like to put my inverters

onto a wood board. This way, if you have to take them out to your car or you have to put them on top of the battery, that negative case of the inverter is not going to touch the positive posts of the battery and short-circuit out and cause a fire. Connecting negative to positive as I told you, is a very bad thing.

First, I will cut my piece of wood a little larger than the inverter. Next, with a power drill, I will screw down the inverter to the piece of wood. This is just a close-up image for your convenience so you can see it's in there nice and tight and snugly screwed down.

We have one inverter mounted on a piece of wood that can now be moved around with ease without short-circuiting any batteries.

I’m going to take it one step further because you might have kids or animals that like to knock things around. But first, I'm going to show you the back of the inverter.

It has two nice large fans on it for keeping it cool. The Whistler is a particularly

intelligent inverter, it only turns on the fans to different speeds as it needs to. Also note the very large negative terminal and the very large positive terminal that we are going to be hooking up the big cables to.

I'm going to get out my ink pen and I'm going to start making some dots in the corners of the piece of wood. I'm going to put one dot in each corner such that it is sitting over a slot in the shelf below it.

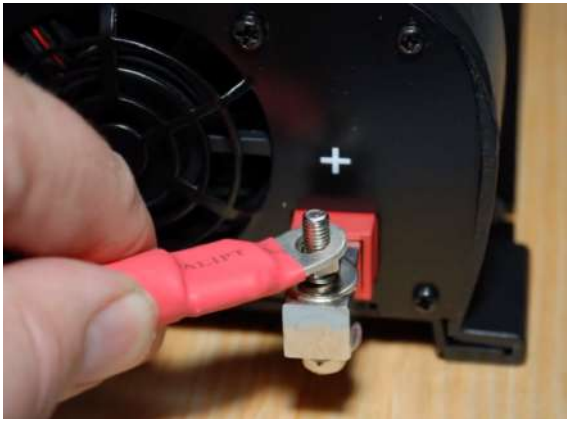
Now, because I'm going to zip tie this down, I need to find a drill bit that is slightly bigger than the zip ties. Then I'm going to drill out all four holes on the dots all the way around the board.

Next, grab your battery cables and take them out of the package because we're going to hook them up.

The cable, both the red and the black for the inverter going to the battery, will have two different size holes on each of the cables.

You'll have one with a bigger hole and you'll have one with a smaller hole.

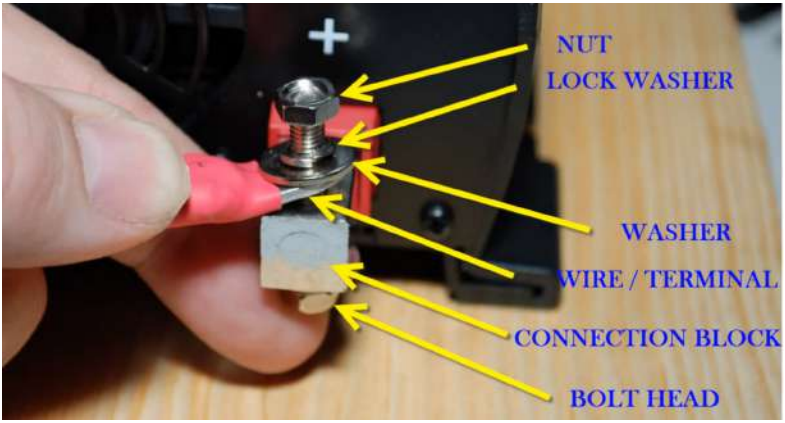
The one with the smaller hole goes over the bolt that is connected to the inverter and the one with the larger hole goes perfectly over the connecting post for the battery.



This should be pretty straightforward for most inverters that we're going to connect the

positive to the positive post. So, take the nut, the lock washer and the washer off the top, but leave the washer and the bolt in as you can see it down there on the bottom.

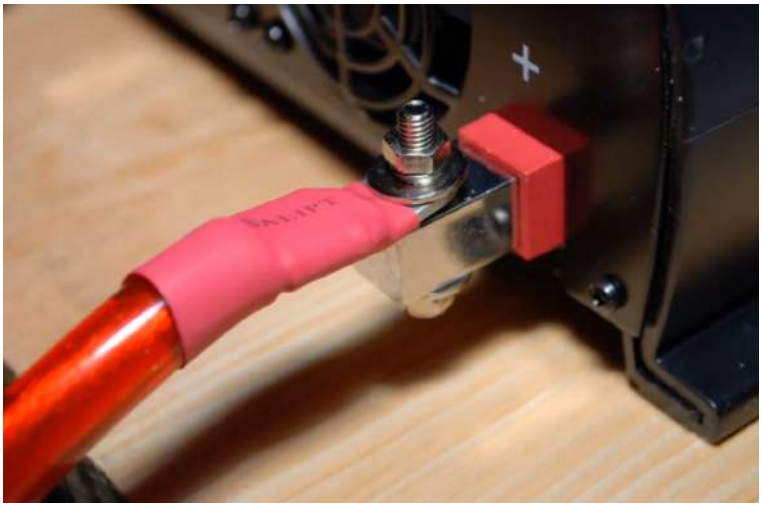
Push up the bolt and slide the battery cable terminal over the bolt up against the large contact block. The battery cable is going to go on first.



On the bottom, we have the head for the bolt, then we have the connection block. All the current is going to flow through this.

Right up against the connection block, we have the terminal for the inverter cable, then the flat washer, the big one, the lock washer on top of the flat washer. Last, but not least, we have the nut on top of that.

Then, screw everything down so it is nice and tight and the cable is going in the direction that will be most convenient for you.



We are going to have a large amount of current going through here, well over 100 amps, so you cannot have a weak connection.

I had to hold the nut head on the bottom of the block and then put the channel locks or players or nut driver on top of the nut on top, to get it to screw down snug, as you can see.

Do everything we just did on the red cable for the black cable. This is what it should look like.

The red and black cable are screwed down tight and the cables are going down through the shelf, running to the golf cart batteries. The red cable coming from the inverter is going to the positive terminal on the battery.

Do exactly the same for the black wire. It goes to the negative terminal of the battery. Just so you get an overall picture of things while we are doing this and progressing, here is an overall photo of what we are doing.

We have two levels of the shelf built, with the inverter sitting on top of the shelf and

we have the two golf cart batteries sitting in front of the shelf at this moment.

Did you notice the way I ran the cables to the golf cart batteries? Notice how I have this cable running off it at about the five o'clock position. I have plenty of room in there between the bolt and the nut and the main battery post.

I did exactly the same thing on the positive post with the cable running completely away from the post. This is so you

have plenty of room to put these terminals on the battery post if you have to.

Let’s say you're using a scavenged car battery after a tornado to hook up your big inverter to the car battery.

If it does not have the nut and the bolt like this battery does, you will need one of these terminal adapters. They clamp on around the post and they give you a great big 3/8-inch bolt to attach to.

Note, the post on some batteries might be a little bigger or smaller whether it's negative or positive. One will go on the positive terminal and one will go on the negative terminal. Just make sure they are tightened up.

Okay, let's slide the golf cart batteries back up on the shelf where they belong. Remember, these things weigh 64 pounds apiece. Tip up one end of the battery, put it over the shelf and then shove and slide it all the way on the shelf so it is centered and sitting nice and cozy.

Do the same thing for the other battery and now your batteries are in their spot where they should be and are ready for use.

Most all inverters made today that are 1000 watts and higher have a display on the front of them. These will usually show you the wattage and the voltage. They'll alternate back and forth between them. A lot of them have a USB plug on them, but funny enough, it's usually a 500 mA USB plug and you really need a thousand or 2000 mA. They also

usually have a spot for a remote jack, this is just a little panel that can go someplace so you can turn the inverter on and off.

If I turn down the lights, you can see the display says the battery is at 12.4 volts and when it switches over to wattage, since there's nothing plugged in to it, it says 00.00.

It's time to tighten down our inverter. Now that we have the batteries in place. Put a cable tie through the hole and bring it back up

through the shelf and then zip tie it and cut off the tail.

Now that your inverter is tied down, it’s nice, safe and secure. No accident will push the inverter off of the shelf.

Since we have a really big inverter and really big batteries, let's look at a really big battery charger.

We are going to take the big clamps off the big battery charger and clamp them onto the big posts of the golf cart battery.

You will note that this charger is a little bit bigger than the other one, you're not going to lift up this one with one hand. This is not one of those old wheel-around chargers, this is made by Schumacher. It is a modern

computer-controlled three-stage battery charger.

It will output up to 55 Amperes at one time into your battery system. That is dumping an awful lot of current. Remember, our other large charger, the one you could pick up with one hand, has a maximum of 30 amps.

What is the advantage of the bigger 55-amp charger? If you are trying to charge up your battery system in a hurry, it’s the best thing.

Let's say you have your little Honda generator outside and you got it turned on and you're running the microwave and doing some other things in the house with the higher power and you want to put some energy into your batteries. This is the time to charge them up and the bigger the charger, the more energy you're going to dump into the first 80% of the capacity of the battery.

Now we have everything hooked up and ready to go, including the charger, the inverter and the batteries.

I'm actually ready to put the other shelves on top of it so I can stock up more emergency preparedness stuff over my battery bank, what better place to have it?

But first, I need to hook up some of my instrumentation. I always like to have a real voltmeter running on all my battery banks. I also love having DC outlets on my battery banks because you never know when the AC will not work or you have a DC only device.

However, we have an inverter tied down to the batteries, we have battery clamps from the charger and now we're putting on more clamps from this. You start getting into a situation where you're putting clamps on top of clamps.

The inverter is a high current device. The battery charger is a significantly high current device, up to 55 amps. So, it has first priority on the big post. That's the big clamp on the big post.

The little 12-volt accessory box I have up there with sockets, is just clamping on to the other clamp. We are nearly ready to put another inverter on here and it's just going to clamp on to the bigger clamp, as well.

You're going to start getting clamp upon clamp upon clamp and I'll show you a little trick in a moment.

So, you have your DC cable, which is a pair of clamps going into one cigarette lighter, then the four-way and one cigarette lighter plug going to four female sockets with the voltmeter.

If you need to plug this into all of the connectors on top of your battery post, what are you to do? Well, what you do is you take your clamp and you get ready to clamp it directly onto the handle of another clamp, see what I did?

I raised up the piece of red insulation which is just covering the metal part of the clamp and I clamped it on. I've had clamp upon clamp upon clamp like this many times, and it works. Remember, this is for a low current device only. The bigger the device, the closer and better connection it has to have directly on the battery.

As I have mentioned previously and will reiterate, you are never connecting anything to the middle posts in between your two 6-volt batteries, it's like they don't exist.

You are only connecting to the big positive post all the way on the left and the big negative post all the way on the right. You might as well just put some duct tape over them so you are reminded to never connect to these.

Now we have our 12 volt 4-way hooked up and I even usually put a zip tie on the cable just so it won't move around, get knocked off or fall down. As you can see, it is nicely lit up and being used.

I have three high current USB chargers plugged into them and they're all glowing blue. These are all 2000 mA chargers and will charge your iPad or anything you want.

I have the voltmeter sitting in there and it says we're charging at 13.76 volts, the green light is on, everything is good.

One little trick I do that I want to show you, is if I want to leave the USB chargers in the device but I don't want them on and running all the time. I just pull them out away from the end of the socket a little bit, they're still in there, it's just that the tip of them is not connected to the back of the socket.

So, when you need them, all you have to do is take your thumb and shove them in and they'll light up, turn blue and are ready for you to use.

I now have another inverter hooked up. This is a 400-watt inverter and is going to be clamping down the batteries.

I generally like to have a smaller wattage inverter for running smaller loads. It is more efficient and my big inverter is off and not being used.

The little inexpensive 400-watt inverter will do just a perfect job of running my little compact fluorescent or LED light that's on all the time and it'll take care of most of the things that I need at this moment.

Keep in mind, everything is a little bit different and, in this case, this is the Duracell 400-watt inverter. You can't just tap the switch to turn it on or turn it off. You have to hold it down for five seconds to turn on the inverter.

Once you do, it'll light up and, as you can see, it is showing us that we are charging at 13.8 volts.

When things are crazy, you will forget the simplest things, so putting labels on your emergency equipment, labeling your boxes, what's in them, putting instructions on stuff, always works very well and it will pay off for you in the long run.

I put a piece of blue tape with marker on it that says ‘Hold down for five seconds’, that way, if I send anyone else to turn it on, they'll see it and they'll hold it down for five seconds.

Here is the result of our final product. I have two golf cart batteries on the shelf on the bottom. You could have 4, 6, or 8, it could be any combination thereof, you can make it as big as you desire.

I took off the 55-amp charger and I put the 30 amp one back on. This is the one I use for my personal use. It works just fine, I don't need to dump the energy into the batteries that fast.

Next to it, I have the 400-watt inverter and a 1600-watt inverter. Above that on the shelf, I have my DC power connector and then I have a shelf on top that I put boxes, tubs and all my other miscellaneous electrical supplies.

I have extension cords and three-way outlets that I can plug into each other. There are little extension cords and extra LED light bulbs and compact fluorescents.

This is what a basic battery system looks like in the basement of the house I currently have and it is growing from here.

Here's another shot of the battery bank, and notice I have a light on. This is a little LED light, about 4 watts. It's on all the time to let me know that everything is working.

Plus, you can see in the upper left-hand corner, the voltmeter with the green light telling me everything is ‘okay’. The 1600-watt inverter is off and the 400-watt inverter is on. The charger is turned on and I have blue tape on both the charger and the inverter to remind

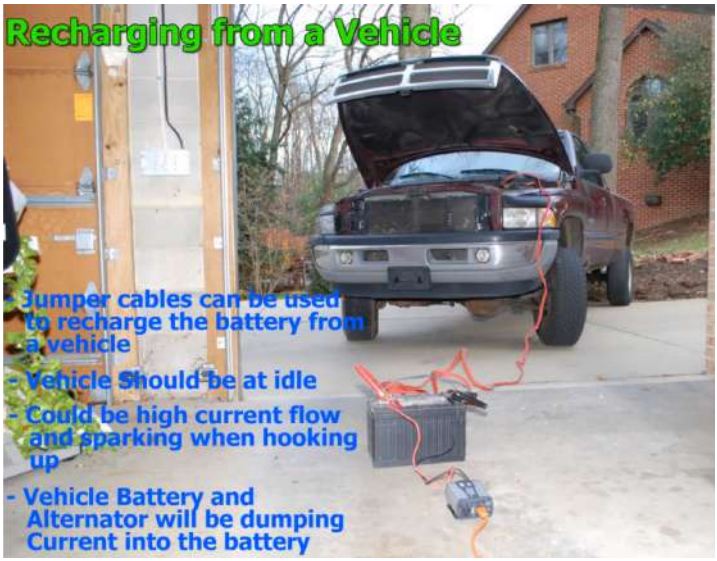
me of all the things I might forget in the heat of the moment.

RECHARGING FROM A VEHICLE

We are going to back up now and we're going to go back to our smaller battery bank and some of the flexibilities and different things you can do to make this thing more useful for you.

If your internal battery inside the house, is becoming low on charge and you want to recharge it from a car, you will need some good long jumper cables.

When you hook up a battery that has been significantly discharged to a fully charged battery in the truck with the alternator running, you can get a very high current dumping into the battery. This is not a three-stage computerized intelligent charger, it is a raw bulk charge dump of current into the battery, so there could be some sparking when you're hooking up.



Clamp on to the battery first and then clamp onto the battery in the truck next. Make sure when you're moving those clamps that they are not touching each other so that they don’t short each other. Otherwise, you're going to have lots of sparks.

Please note, like I said, this is not a computer-controlled three-stage intelligent charger. This is what we would call the bulk charge stage. You're going to charge up the battery pretty quickly, as fast as you can, up to about 80% of its charge. It is not the gentlest thing on your battery, but it will work just fine.

Remember, the battery is there to be used, so use it. While I am charging the battery and running the truck, the inverter can still be hooked up to this at the same time. You can hook it up to this battery or you can hook it up to the battery in the truck.

If you are going to power your refrigerator or freezer, at these higher loads, you especially want to clamp on and run it to the car, start the car and run it at idle.

Even if you're going to run your coffee maker, because your coffee maker can be around 1100 watts, that could be 3, 4, 5 times the amount of current your refrigerator is drawing.

I had lots of people write to me about running their coffee maker off their inverter off of the car. They've even made mention that other people were waiting in line at Starbucks for over an hour for a cup of coffee during hurricane Sandy.

Now, coffee pots are not low power devices, they can draw between 1100 and 1500 watts. It is a good idea to run one with the car at idle and then let it idle for another 20 minutes after you turn off the pot.

If your pot has a keep hot feature where it keeps the pot hot after you brew it, unplug it. Try to only brew one pot of coffee per hour.

You should have a 1500 watt plus inverter to do this. Don't power the refrigerator or anything else while you are brewing the coffee. Some coffee pots take 800 watts, the one you see here is 1100 watts.

- Coffee pots are NOT low power devices. They can draw 1100 to 1500 watts.
- Good idea to run one with the car at idle and then to let it idle for another 20 minutes
- Try to only brew 1 pot of coffee per hour from your car at idle and the battery and the inverter.
- You should have a 1500 watt+ inverter
- Don't power the refrigerator or anything else while you are brewing the coffee.

I've seen others that were 1400 watts, but to do this, you really should have a good 1500-watt inverter, and again, because you have a 1500-watt inverter, it doesn't mean you can draw power all the time at this level. All it means is you're going to draw 1500 watts for a short period of time.

Just a little side fun note for you: If a coffee pot draws 1100 watts of power and it takes 10 minutes to make a pot of coffee, that is the exact same amount of battery energy that would run this 2-watt LED bulb for 91 hours.

So, it's either one pot of coffee in 10 minutes or 8 days of light. That is a lot of energy to be drawing off of a battery to make a pot of coffee.

Again, since you are charging your battery while at low charge and the battery in your truck is at high charge along with the alternator, it is going to dump a lot of current really quickly into the battery.

And since its high current, you want to make sure you have very secure connections to the vehicle battery from the clamps.

Make sure they're on tight, wiggle them back and forth so the teeth cut into the lead of

the battery terminals and make sure you have a really good connection when you are doing this.

OTHER EXAMPLES OF BATTERY SYSTEMS

Here are some photographs from some other folk's battery systems. This is a battery system from a friend of mine that has 8 full size GC2 golf cart marine batteries in this system. This would equal about 600 Ah of power, which is a lot.

The battery system is split in 2 for charging so that it will charge quicker. You can use the charger on the left to charge the left-hand side and the charger on the right to charge the right-hand side.

This will keep everything charged so that you can throw one big switch joining both of the banks together for better efficiency to run the inverters.

Notice on the bottom right-hand side of the batteries, there is a specific gravity battery hydrometer. It is taped right to the battery so the health of the batteries can be checked instantaneously.

With this battery system, a wooden shelf was built to hold all the batteries and you will notice under the shelf is where the inverters

are mounted, that way they get plenty of air. There is even a little fan in there just to blow more air onto the 1500-watt inverter.

In fact, there is a 1500-watt inverter, a 400-watt inverter and a 175-watt inverter. All hooked up to the same battery system. The more you have, the better.

Now, this battery bank is built into a little closet in the garage, hidden away, so there are two voltmeters sitting outside of the closet going through a hole in the wall, with a 12-volt socket.

Remember, I told you the battery system is split into two for purposes of charging quickly, so one voltmeter is on one half of the battery system and another on the other half.

Here is one more battery system example for you.

You can see we have two batteries. They are actually stacked vertically and are connected together just like when our batteries are sitting next to each other.

We have two marine batteries and the battery chargers are just clamped on like you would in any other situation. There were no wires cut or eyelets put on.

The battery chargers were kept flexible so they could be taken out and used in a car for instance. They are just clamped onto the terminals of the battery and will stay there and continue to work.

There are also two inverters hooked up to both the batteries. These are both 1200-watt inverters and either one can be used or they can both be used at the same time. When they are not being used, as they are right now, they are both turned off.

Here we have two battery chargers, both hooked up to the batteries. These are my old favorite, the Vector chargers that are no longer available. Notice how he has two chargers. The top one is on, it reads ‘FUL’ and is always maintaining the battery charge level.

The bottom one is off and unplugged. So, there are two batteries, two inverters, and two chargers. Two is one, one is none.

This concludes the emergency home power system and the building of it. By now,

you should have a PhD in building a system for your own family.

I really hope I have covered enough information and detail to give you the options and ideas on what you can do. I encourage you to give the battery system a try because you never know when, why and how long your electricity is going to go out.

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